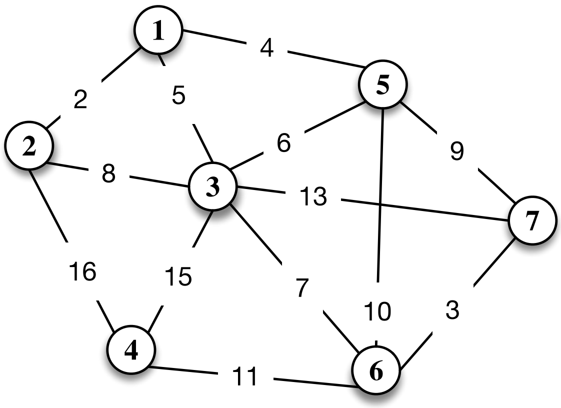
NOME: COGNOME: MATRICOLA: Exam 11/2/2020

questa è stata l'ultima che ho visto
devo andare indietro nel tempo

First part : Closed book

Total score: 21 pts. Minimum score to pass: 11 pts. Tentative scores: 1) 4 pts, 2) 5 pts, 3) 5 pts, 4) 3 pts.

1. Consider the undirected graph in the figure below, with costs reported on the arcs. Compute the minimum cost spanning tree illustrating the intermediate steps.



Once the solution has been found, determine for arcs (5, 7) and (3, 6), separately, what is the cost coefficient interval that guarantees that the optimal solution remains unchanged. Justify the answer.

2. Consider the following linear programming problem:

$$\begin{aligned} \max \quad & -x_1 - x_2 \\ & x_1 + 3x_2 \leq -3 \\ & -2x_1 + x_2 \leq 0 \\ & x_1 - 4x_2 \leq 12 \end{aligned}$$

Is $x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ a feasible solution? Write the auxiliary problem to determine a starting feasible solution. What is a trivial feasible solution for the auxiliary problem? Provide the set of active constraints for the found feasible solution in the auxiliary problem, and the conditions for the existence of a growing feasible solution.

3. Consider the following integer linear programming problem:

$$\begin{aligned} \min \quad & 5x_1 - x_2 \\ & 3x_1 + 3x_2 \quad \leq 13 \\ & -8x_1 + 3x_2 \quad \leq 8 \\ & x_1, x_2 \in \mathcal{Z} \end{aligned}$$

Add the slack variables (x_3 and x_4) to the first and the second constraints, and consider the optimal base $B = \{2, 3\}$ of the LP relaxation. Generate the Gomory cuts associated with the fractional components of the solution. Justify your answer and give the geometric representation of the feasible region and of the cuts.

4. Consider the optimization problems of the form $P : \max\{cx : x \in F\}$ and $P' : \max\{cx : x \in F'\}$. What are the characteristics of F' such that P' is a relaxation of P? Let x' be the optimal solution of P' and x^* be the optimal solution of P. Can we always say that $cx' \geq cx^*$? Why? Let us suppose that P' does not admit any feasible solution, what can we say of P?

AMPL questions: 4 points. **in the rear of this paper**

Write the definition in the AMPL modeling language of the following:

- a parameter n
- a set `I` of all the integers between 1 and n.
- a set of positive variables x_i with $i \in \{I \cup 0\}$.
- a set of parameters b_i with $i \in I$.

Then using the defined variables and parameters write in the AMPL modeling language the following constraint:

$$x_{i-1} \leq b_i x_i \qquad \forall i \in I : i \leq 5$$

ANSWER IN THE SPACE BELOW